

Wherever you see **red characters**, these need to be replaced by your information. This includes the **< >** symbols!

Scatterplot of Relationship between Two Quantitative Variables

```
ggplot(<NAME OF DATASET>,  
      aes(x = <NAME OF EV>,  
          y = <NAME OF RV>)) +  
  geom_point()
```

Creates a scatterplot with EV on the x-axis and RV on the y-axis.

Scatterplot of Relationship between Two Quantitative Variables WITH Simple Linear Regression Line

```
ggplot(<NAME OF DATASET>,  
      aes(x = <NAME OF EV>,  
          y = <NAME OF RV>)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = F)
```

Creates a scatterplot with EV on the x-axis and RV on the y-axis along with the simple linear regression line

Scatterplot of Relationship between Two Quantitative Variables WITH Color & Shape by Third Categorical Variable

```
ggplot(<NAME OF DATASET>,  
      aes(x = <NAME OF QUANTITATIVE EV>,  
          y = <NAME OF RV>)) +  
  geom_point(aes(shape = <NAME OF CATEGORICAL EV>,  
                 color = <NAME OF CATEGORICAL EV>))
```

Creates a scatterplot with a quantitative EV on the x-axis and RV on the y-axis, with the colors and shapes of points corresponding to a categorical EV.

Fit a Regression Model and See Model Output

```
mymod <- lm(<NAME OF RV> ~ <NAME OF EV1> + <NAME OF EV2> + ...,  
            data = <NAME OF DATASET>)  
summary(mymod)
```

Output will be the estimated coefficients for the model and other information like R^2 etc.

Confidence Intervals for Regression Coefficients

Assuming you have fit a model like above called **mymod**

```
confint(mymod, level = 0.95)
```

Output will be the lower and upper bounds for a 95% confidence interval for every coefficient. You can change the level to get different X% confidence intervals.

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Scatterplot of Residuals versus Predicted Values

Assuming you have fit a model like above using `lm()` called **mymod**

```
moddat <- mymod |>
  augment()

ggplot(moddat) +
  geom_hline(yintercept = 0, color = "blue") +
  geom_point(aes(x = .fitted, y = .resid)) +
  labs(x = "Predicted Values",
       y = "Residual")
```

Creates a plot of residuals versus predicted values from a model with a blue line at 0.

Scatterplot of Relationship between Two Quantitative Variables WITH Simple Linear Regression Line

Assuming you have fit a model like above using `lm()` called **mymod**, and saved a dataset using `augment()` like **moddat** above:

```
ggplot(moddat, aes(x = .resid)) +
  geom_histogram() +
  labs(x = "Residual",
       y = "Count",
       title = "Distribution of Residuals")
```

Creates a histogram of residuals from a model.